Agent-Based Modeling and Simulation

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Dibris

also based on tutorials by Stephany Coffman-Wolph, Steve Kaisler, Greg Madey

Agent-based simulation

C. M. Macal and M. J. North, Proc. of Winter Simulation Conference, 2007

"... Agent-based modeling and simulation (ABMS) is a new approach to modeling systems comprised of autonomous, interacting agents. ABMS promises to have far-reaching effects on the way that businesses use computers to support decision-making and researchers use electronic laboratories to support their research."

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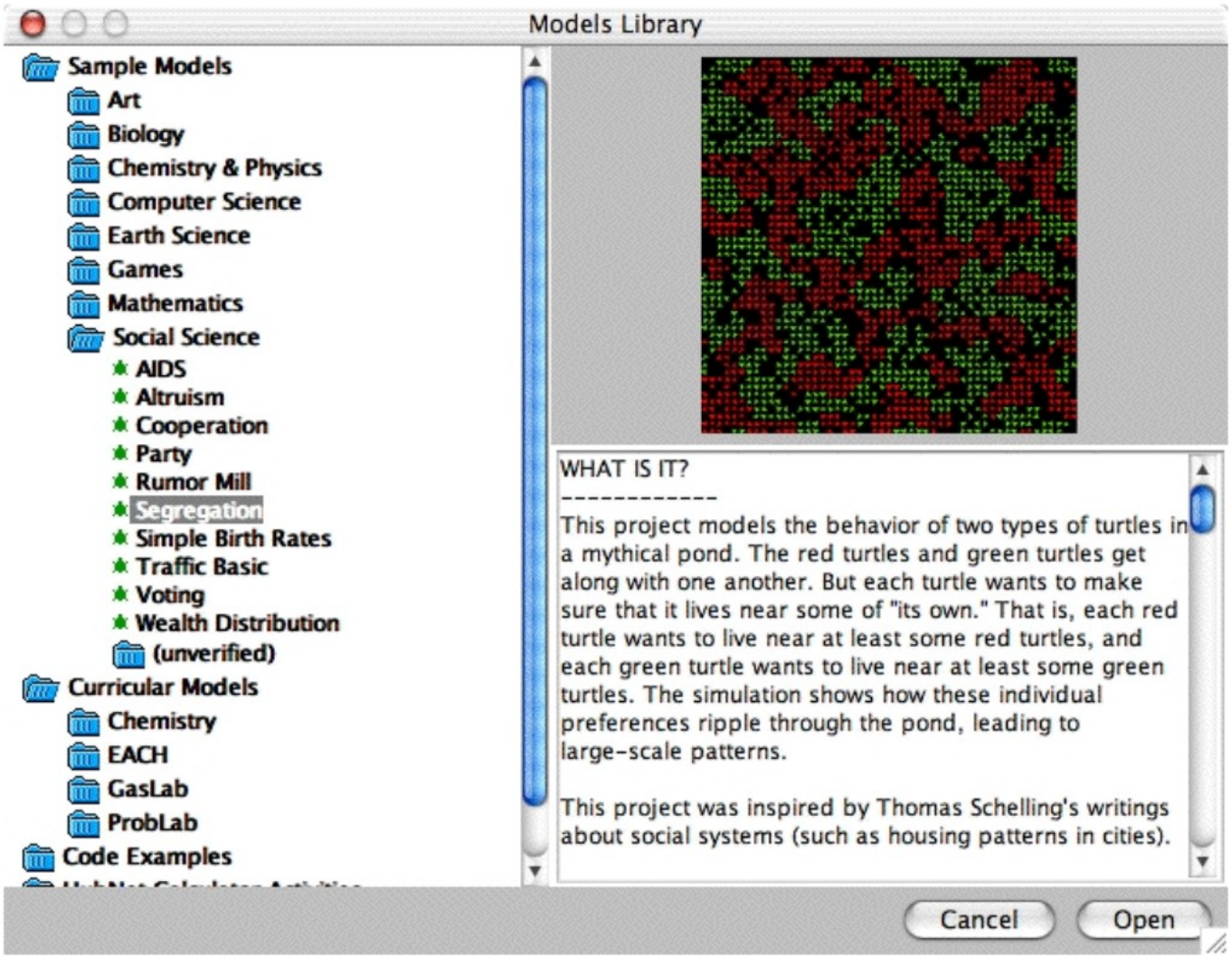
Agent-based simulation

C. M. Macal and M. J. North, Proc. of Winter Simulation Conference, 2007

"Some have gone so far as to contend that ABMS "is a third way of doing science," in addition to traditional deductive and inductive reasoning.
Computational advances have made possible a growing number of agent-based models across a variety of application domains.
Applications range from modeling agent behavior in the stock market, supply chains, and consumer markets, to predicting the spread of epidemics, the threat of bio-warfare, and the factors responsible for the fall of ancient civilizations. "

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Agent-based simulation: how?



NetLogo, `http://ccl.northwestern.edu/netlogo/`

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WHAT IS A SWARM?

- A loosely structured collection of interacting agents
  - Agents:
    - Individuals that belong to a group (but are not necessarily identical)
    - They contribute to and benefit from the group
    - They can recognize, communicate, and/or interact with each other
- The instinctive perception of swarms is a group of agents in motion – but that does not always have to be the case.
- A swarm is better understood if thought of as agents exhibiting a collective behavior

EXAMPLES OF SWARMS IN NATURE

Classic Example: Swarm of Bees



Vitellogenin-rich bees rearing swarm cells. R. Oliver

EXAMPLES OF SWARMS IN NATURE

Can be extended to other similar systems:

- Ant colony
  - Agents: ants



EXAMPLES OF SWARMS IN NATURE

Can be extended to other similar systems:

- Flock of birds
  - Agents: birds



EXAMPLES OF SWARMS IN NATURE

Can be extended to other similar systems:

- School of fish
  - Agents: fish



EXAMPLES OF SWARMS IN NATURE

Can be extended to other similar systems:

- Traffic
  - Agents: cars



EXAMPLES OF SWARMS IN NATURE

Can be extended to other similar systems:

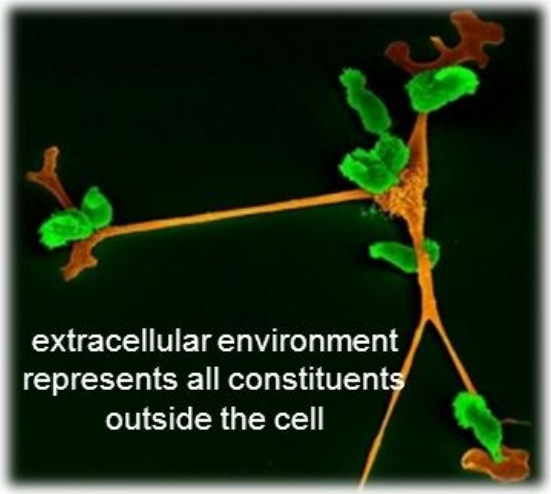
- Crowd
  - Agents: humans



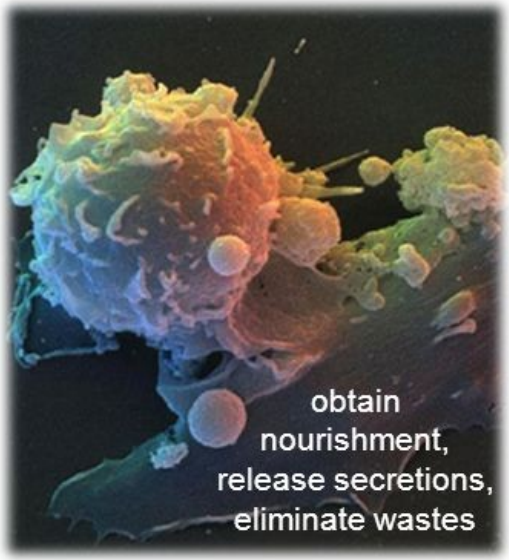
EXAMPLES OF SWARMS IN NATURE

Can be extended to other similar systems:

- Immune system
  - Agents: cells and molecules



CHEMICAL REGULATORS



impact/interact neighboring (1) cells,
(2) different tissues &/or
(3) different organs

EXAMPLES OF SWARMS IN NATURE

Can be extended to other similar systems:

– ...the mind?



We shall envision the mind (or brain) as composed of many partially autonomous "agents" - as a "Society" of smaller minds. ...It is easiest to think about partial states that constrain only agents within a single Division. ...(we suggest) the local mechanisms for resolving conflicts could be the precursors of what we know later as reasoning - useful ways to combine different fragments of knowledge.

M. Minsky, K-Lines: A Theory of Memory (1980)

SWARM INTELLIGENCE

Swarm intelligence (SI) is the collective behavior of decentralized, self-organized systems, natural or artificial. The concept is employed in work on artificial intelligence. The expression was introduced by Gerardo Beni and Jing Wang in 1989, in the context of cellular robotic systems.

[Beni, G., Wang, J. Swarm Intelligence in Cellular Robotic Systems, Proceed. NATO Advanced Workshop on Robots and Biological Systems, Tuscany, Italy, June 26–30 (1989)]

SWARM INTELLIGENCE

The concept of SI was expanded by Bonabeau, Dorigo, and Theraulaz in 1999 (and is widely recognized by the scientific community):

“Using the expression ‘swarm intelligence’ to describe only this work seems unnecessarily restrictive: that is why we extend its definition to include devices inspired by the collective behavior of insect colonies and other animal societies”

[Swarm Intelligence: From Natural to Artificial Systems
Eric Bonabeau, Marco Dorigo and Guy Theraulaz
New York, NY: Oxford University Press, Santa Fe Institute Studies in the Sciences of Complexity
(1999)]

SWARM INTELLIGENCE

SI systems consist typically of a population of simple agents or boids* interacting locally with one another and with their environment. The inspiration often comes from nature, especially biological systems. The agents follow very simple rules, and although there is no centralized control structure dictating how individual agents should behave, local, and to a certain degree random, interactions between such agents lead to the emergence of "intelligent" global behavior, unknown to the individual agents.

* Boids is an artificial life program, developed by Craig Reynolds in 1986, which simulates the flocking behaviour of birds. His paper on this topic was published in 1987 in the proceedings of the ACM SIGGRAPH conference. The name "boid" corresponds to a shortened version of "bird-oid object", which refers to a bird-like object.



ADVANTAGES OF SWARM INTELLIGENCE

- The systems are scalable because the same control architecture can be applied to a couple of agents or thousands of agents
- The systems are flexible because agents can be easily added or removed without influencing the structure
- The systems are robust because agents are simple in design, the reliance on individual agents is small, and failure of a single agents has little impact on the system’s performance

AGENT BASED MODELING AND SIMULATION

- Agent-based modeling is a computational methodology that allows scientists to create, analyze, and experiment with artificial worlds populated by agents that interact in nontrivial ways and that constitute their own environment.
- An agent-based simulation (ABS) is one in which a collection of agents compete or collaborate or both to attain individual goals in pursuit of a social or organizational goals.
- A simulation based on software agents can support good science provided the design of the agents and the simulation environment are themselves based on good science.

WHY ABMS?

- Aspects of the real world that we want to examine are not physically accessible
- ABMS takes place in an artificial world
- Experimenting with the real system is prohibited due to undesirable disturbances
- Time scale of system behavior is too small or too large for observation
- The original system does not exist any more or does not exist yet
- Modeling is a tool for understanding formalization of a hypothesis that otherwise would have remained very vague

ABMS APPLICATIONS

Biology, energy, transportation, entertainment...

https://scholar.google.it/scholar?as_ylo=2020&q=applications+of+agent+based+modelling+and+simulation&hl=en&as_sdt=0,5&as_rr=1&as_vis=1

https://www.youtube.com/watch?v=042YjQ9aZNk

https://www.massivesoftware.com/

NETLOGO

